Santa Clara, CA, 95050
mehrab-tanjim.github.io
linkedin.com/in/mehrab-tanjim

# Mehrab Tanjim

(858) 214-9372
mehrabtanjim@gmail.com
tanjim@adobe.com

## SUMMARY

Highly accomplished researcher and industry professional with 7+ years of experience in developing and deploying scalable machine learning solutions in various fields such as query understanding and disambiguation, mitigating biases in generative models, and designing recommender systems. Currently focusing on automated optimization of LLM-based agents for enhancing reasoning, planning, and decision-making. Proven track record of publishing research papers in top-tier conferences and issued patents in AI and machine learning.

## EDUCATION

**PhD in CS – University of California, San Diego (UCSD)** *Sep. 2018 – Mar. 2023*

- Thesis Research Focus: Debiasing Generative Networks

**MS in CS – University of California, San Diego (UCSD)** *Sep. 2018 – March 2021*

- Coursework: Neural Networks/Pattern Recognition, Deep Learning for Sequences, Deep Unsupervised Learning, Computer Vision, Convex Optimization

**BS in CS – Bangladesh University of Engineering and Technology (BUET)** Apr. *2012 – Feb. 2017*

- Coursework: Database, Operating System, Computer Networks, Software Development, Artificial Intelligence, Simulation and Modelling, Machine Learning, Pattern Recognition

## EXPERIENCE

**Adobe Research** **San Jose, CA, USA**
*Research Scientist II (Full Time)* *Apr. 2023 – Present*

- **Automated Optimization of LLM-based Agents:** Leading the development of automated optimization techniques for LLM-based agents, improving the performance to reason, plan, and make decisions.
- **Synthetic Data Generation for AI Model Optimization**: Designed and implemented a synthetic data generation framework to mitigate data scarcity challenges in AI model optimization and evaluation phases.
- **Research on Query Understanding and Disambiguation:** Led research on query understanding and disambiguation, exploring novel techniques to improve the accuracy and robustness of AI models in handling complex and ambiguous queries. My solution has been productionized in making Adobe AEP AI Assistant. Mainly my solutions can be categorized into three parts: 1. Ambiguity Detection **[Pdf]**, 2. Query Rewriting **[Pdf]**, and 3. Asking Clarifying Questions **[Pdf]**.

**The Home Depot** **Atlanta, GA, USA**
*Data Science Intern* *Sep 2022 – Dec. 2022*

- **Subsequent Purchase Prediction:** Improved the current model for subsequent purchase prediction by 1.5% by using an autoregressive model and sampled embedding loss which can further be used to generate synthetic datapoints, denoise training data and recommend new items using metadata embedding **[Patent].**

**Adobe Research** **San Jose, CA, USA**
*Research Scientist Intern* *June 2022 – Sep. 2022*

- **Scalable Video Fingerprinting:** Built a scalable, end-to-end pipeline using FAISS library that can trace a manipulated video in less than a second from a trusted database with millions of corpuses **[Patent]**.

**Adobe Research** **San Jose, CA, USA**
*Computer Vision, Imaging & Video Intern* *June 2021 – Sep. 2021*

- **Debiasing Image-to-Image Translation:** Pretrained StyleGAN2 based networks show various biases in different image-to-image translation tasks (such as super-resolution, sketch-to-image, etc.). Mitigated this bias issue using contrastive learning and uniform sampling of minority attributes **[Patent]**.

**Adobe Research** **San Jose, CA, USA**
*Data Science Intern* *June 2020 – Sep. 2020*

- **Bias Detection and Mitigation:** Identified the bias issue in the image results for search queries, proposed a way to audit. In addition, proposed an attribute-controlled style-based generator to create new content to mitigate such biases and enrich user experience **[Pdf] [Patent]**.

**Etsy, Inc.** **New York, NY, USA**
*Data Science Intern* *June 2019 – Sep. 2019*

- **Intent Detection for Recommendation:** Captured users' hidden intents (i.e. explore, purchase) from their interactions by designing a hierarchical Transformer model (improved recommendations by **5%, accepted in WWW'20**) **[Pdf][Code]**.

**University of California, San Diego** **San Diego, CA, USA**
*Graduate Student Researcher* *Sep. 2018 – Mar. 2023*

- **Dynamic Convolution:** Built an adaptive convolution network which changes its kernel dynamically depending on the current input (~**10%** better recommendations, accepted in **CIKM'20**) **[Pdf] [Code]**.
- **Visual Commonsense Reasoning:** Enforced reasoning for ans. prediction on VCR by building a differentiable module which jointly trains ans. and rationale prediction (performed better in leaderboard) **[Pdf] [Code]**.
- **CNN for Sequences:** Improved the scalability of sequential recommender methods by modelling a scalable depth-wise separable 1D convolution neural network (requires ~**30%** less memory) **[Pdf]**.
- **Rationale Generation:** Tasked state-of-the-art Visual Question Answering model (ViLBERT) with rationale generation (using GPT-2) to interpret/justify answer prediction. It improves accuracy by **1.5%** as well **[Pdf]**.

*Graduate Teaching Assistant* *Fall'19,'20, Winter'20,'22, Spring'20,'21*

- **Neural Networks/Pattern Recognition:** Designed and assessed assignments on DNN, CNN (Image Segmentations), and RNN/LSTM (Image Captioning). Responsible for mentoring deep learning projects.

**BUET** **Dhaka, Bangladesh**
*Research Assistant* *Oct. 2017 – Aug. 2018*

- **Scalable Machine Learning:** Improved the scalability of PCA for large datasets (up to **83×** better performance) using sketching technique **[Pdf] [Code]**.
- **Distributed Algorithm Design/Federated Learning:** Extended both Spark and Hadoop for creating geo-distributed clusters in AWS and designed geo-distributed algorithms for higher dimension data **[Code]**.

## ISSUED PATENTS

• **Scalable Video Fingerprinting for Content Authenticity**, R. Sinha, V. Swaminathan, S. Jenni, *Mehrab Tanjim*, J. Collomosse, United States Patent Application 18/473045, 27 March 2025

• **ENHANCING NEXT ITEM RECOMMENDATION THROUGH CROSS-ATTENTION**, W. Shalaby, X. Cui, J. Balaji, *Mehrab Tanjim*, United States Patent Application 18/736,371, 12 December 2024

• **System and methods for diversity auditing**, *Mehrab Tanjim*, R. Sinha, M. Sinha, D. Arbour, S. Mahadevan, United States Patent Application 17/652,026, 3 December 2024

• **Debiasing image to image translation models**, *Mehrab Tanjim*, K. K. Singh, K. Kafle, R. Sinha, United States Patent Application 17/880,120, 8 February 2024

• **GENERATING SIMULATED IMAGES THAT ENHANCE SOCIO-DEMOGRAPHIC DIVERSITY**, *Mehrab Tanjim*, R. Sinha, K. K. Singh, S. Mahadevan, D. Arbour, M. Sinha, United States Patent Application 17/485780, 2023

## SELECTED PUBLICATIONS

• **Detecting Ambiguities to Guide Query Rewrite for Robust Conversations in Enterprise AI Assistants**, *Mehrab Tanjim*, X. Chen, V. S. Bursztyn, U. Bhattacharya, T. Mai, V. Muppala, arXiv preprint arXiv:2502.00537, 2025

• **Disambiguation in Conversational Question Answering in the Era of LLM: A Survey**, *Mehrab Tanjim*, Y. In, X. Chen, V. S. Bursztyn, R. A. Rossi, S. Kim, G. J. Ren, arXiv preprint arXiv:2505.12543, 2025

• **Exploring Rewriting Approaches for Different Conversational Tasks**, *Mehrab Tanjim*, R. A. Rossi, M. Rimer, X. Chen, S. Kim, V. Muppala, T. Yu, Z. Hu, arXiv preprint arXiv:2502.18860, 2025

• **Calibrating MLLM-as-a-judge via Multimodal Bayesian Prompt Ensembles**, E. Slyman, *Mehrab Tanjim*, K. Kafle, S. Lee, ICCV, 2025

• **SKALD: Learning-Based Shot Assembly for Coherent Multi-Shot Video Creation**, C. Y. Lu, *Mehrab Tanjim*, I. Dasgupta, S. Sarkhel, G. Wu, S. Mitra, S. Chaterji, ICCV, 2025

• **VISIAR: Empower MLLM for Visual Story Ideation**, Z. Xia, S. Sarkhel, *Mehrab Tanjim*, S. Petrangeli, I. Dasgupta, Y. Chen, J. Xu, D. Liu, S. Mitra, D. N. Metaxas, ACL, 2025

• **Diversify-verify-adapt: Efficient and Robust Retrieval-Augmented Ambiguous Question Answering**, Y. In, S. Kim, R. A. Rossi, *Mehrab Tanjim*, T. Yu, R. Sinha, C. Park, NAACL, 2025

• **Self-debiasing large language models: Zero-shot recognition and reduction of stereotypes**, I. O. Gallegos, R. A. Rossi, J. Barrow, *Mehrab Tanjim*, T. Yu, H. Deilamsalehy, R. Zhang, S. Kim, F. Dernoncourt, NAACL, 2025

• **ECLAIR: Enhanced Clarification for Interactive Responses**, J. Murzaku, Z. Liu, *Mehrab Tanjim*, V. Muppala, X. Chen, Y. Li, IAAI 2025

• **RECON: Training-Free Acceleration for Text-to-Image Synthesis with Retrieval of Concept Prompt Trajectories**, C. Y. Lu, S. Agarwal, *Mehrab Tanjim*, K. Mahadik, A. Rao, S. Mitra, S. K. Saini, S. Bagchi, S. Chaterji, ECCV, 2024

• **Discovering and Mitigating Biases in CLIP-based Text-to-Image Generation**, *Mehrab Tanjim*, K. K. Singh, K. Kafle, R. Sinha, G. W. Cottrell, WACV, 2024

• **Debiasing Image-to-Image Translation Models**, *Mehrab Tanjim*, K. K. Singh, K. Kafle, R. Sinha, G. W. Cottrell, British Machine Vision Conference (BMVC), 2022

• **Generating and Controlling Diversity in Image Search**, *Mehrab Tanjim*, R. Sinha, K. K. Singh, S. Mahadevan, D. Arbour, M. Sinha, G. W. Cottrell, Winter Conference on Applications of Computer Vision (WACV), 2022

• **DynamicRec: A Dynamic Convolutional Network for Next Item Recommendation**, *Mehrab Tanjim*, Hammad A. Ayyubi, G. W. Cottrell, Conference on Information and Knowledge Management (CIKM), 2020

• **Attentive Sequential Models of Latent Intent for Next Item Recommendation**, *Mehrab Tanjim*, C. Su, E. Benjamin, D. Hu, L. Hong, J. McAuley, The Web Conference (WWW), 2020

• **Fast, Scalable and Geo-Distributed PCA for Big Data Analytics**, T. M. Tariq Adnan, *Mehrab Tanjim*, M. Abdullah Adnan, Information Systems, Elsevier, 2021

• **sSketch: A Scalable Sketching Technique for PCA in the Cloud**, *Mehrab Tanjim*, Muhammad Abdullah Adnan, Web Search and Data Mining (WSDM), 2018